

- 22** The sequence of bases in mRNA for the first eight amino acids in the β -polypeptide of adult haemoglobin is:

GUG–CAC–CUG–ACU–CCU–GAG–GAG–AAG.

In haemoglobin C, which is a cause of haemolytic anaemia, the sequence is:

GUG–CAC–CUG–ACU–CCU–AAG–GAG–AAG.

The coding for seven of the amino acids is listed.

amino acid	DNA triplet
glu	CTC
his	GTG
leu	GAG
lys	TTC
pro	GGA
thr	TGA
phe	AAG

Which change occurs to the amino acid sequence of adult haemoglobin to make haemoglobin C?

- A** Histidine is changed to leucine.
- B** Proline is changed to threonine.
- C** Glutamic acid is changed to lysine.
- D** Leucine is changed to phenylalanine.